```
# 🍌 Banana Ripeness CNN + Advisory System (Google Drive Version)

!pip install tensorflow pillow numpy matplotlib

import tensorflow as tf
from tensorflow.keras import layers, models
import numpy as np
import matplotlib.pyplot as plt
import zipfile
from tensorflow.keras.preprocessing.image import ImageDataGenerator, load_img, img_to_array
```

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```

```
# 🔗 Mount Google Drive
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
zip_path = "/content/drive/MyDrive/dataset.zip"
```

```
# 📦 Unzip dataset
with zipfile.ZipFile(zip_path, 'r') as zip_ref:
    zip_ref.extractall("data")

train_dir = "data/dataset"
```

```
# 🔄 Data Preprocessing
datagen = ImageDataGenerator(rescale=1./255, validation_split=0.2)
```

```
train_data = datagen.flow_from_directory(
    train_dir,
    target_size=(128,128),
    batch_size=32,
    class_mode='categorical',
    subset='training'
)

val_data = datagen.flow_from_directory(
    train_dir,
    target_size=(128,128),
    batch_size=32,
    class_mode='categorical',
    subset='validation'
)
```

```
Found 454 images belonging to 4 classes.
Found 112 images belonging to 4 classes.
```

```
# 🧠 CNN Model
model = models.Sequential([
    layers.Conv2D(32, (3,3), activation='relu', input_shape=(128,128,3)),
    layers.MaxPooling2D(2,2),

    layers.Conv2D(64, (3,3), activation='relu'),
    layers.MaxPooling2D(2,2),

    layers.Conv2D(128, (3,3), activation='relu'),
    layers.MaxPooling2D(2,2),

    layers.Flatten(),
    layers.Dense(128, activation='relu'),
    layers.Dropout(0.5),
    layers.Dense(4, activation='softmax')
])
```

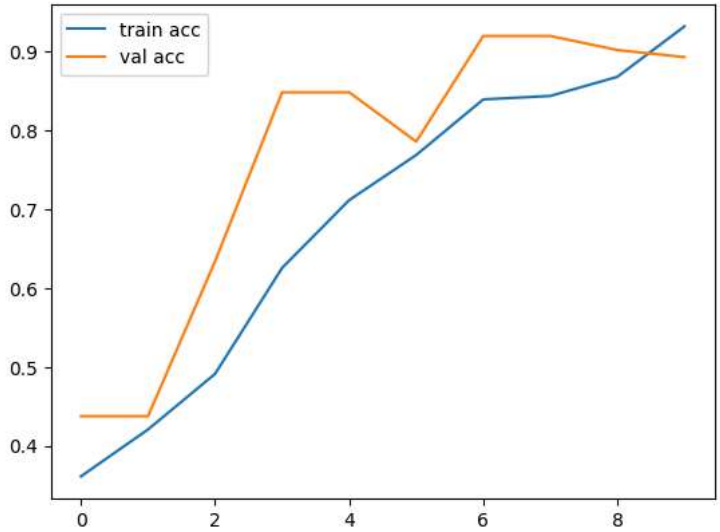
```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/convolutional/base_conv.py:113: UserWarning: Do not pass an `input_shap
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

```
# ⚙️ Compile Model
model.compile(optimizer='adam', loss='categorical_crossentropy', metrics=['accuracy'])
```

```
# 🏋️ Train Model
history = model.fit(train_data, validation_data=val_data, epochs=10)
```

```
Epoch 1/10
15/15 ━━━━━━━━━━━━━━━━━━━━ 41s 3s/step - accuracy: 0.3612 - loss: 1.4160 - val_accuracy: 0.4375 - val_loss: 1.2703
Epoch 2/10
15/15 ━━━━━━━━━━━━━━━━━━━━ 35s 2s/step - accuracy: 0.4207 - loss: 1.2599 - val_accuracy: 0.4375 - val_loss: 1.1854
Epoch 3/10
15/15 ━━━━━━━━━━━━━━━━━━━━ 30s 2s/step - accuracy: 0.4912 - loss: 1.1193 - val_accuracy: 0.6339 - val_loss: 0.8573
Epoch 4/10
15/15 ━━━━━━━━━━━━━━━━━━━━ 30s 2s/step - accuracy: 0.6256 - loss: 0.8902 - val_accuracy: 0.8482 - val_loss: 0.6269
Epoch 5/10
15/15 ━━━━━━━━━━━━━━━━━━━━ 29s 2s/step - accuracy: 0.7115 - loss: 0.7335 - val_accuracy: 0.8482 - val_loss: 0.4773
Epoch 6/10
15/15 ━━━━━━━━━━━━━━━━━━━━ 29s 2s/step - accuracy: 0.7687 - loss: 0.6138 - val_accuracy: 0.7857 - val_loss: 0.5030
Epoch 7/10
15/15 ━━━━━━━━━━━━━━━━━━━━ 41s 2s/step - accuracy: 0.8392 - loss: 0.4216 - val_accuracy: 0.9196 - val_loss: 0.2674
Epoch 8/10
15/15 ━━━━━━━━━━━━━━━━━━━━ 41s 2s/step - accuracy: 0.8436 - loss: 0.4145 - val_accuracy: 0.9196 - val_loss: 0.2410
Epoch 9/10
15/15 ━━━━━━━━━━━━━━━━━━━━ 28s 2s/step - accuracy: 0.8678 - loss: 0.3333 - val_accuracy: 0.9018 - val_loss: 0.4217
Epoch 10/10
15/15 ━━━━━━━━━━━━━━━━━━━━ 29s 2s/step - accuracy: 0.9317 - loss: 0.2144 - val_accuracy: 0.8929 - val_loss: 0.3265
```

```
# 📊 Plot Accuracy
plt.plot(history.history['accuracy'], label='train acc')
plt.plot(history.history['val_accuracy'], label='val acc')
plt.legend()
plt.show()
```




```
# 💾 Save Model to Drive
model.save("/content/drive/MyDrive/banana_model.h5")
```

```
WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save_model(model)`. This file format
```

```
def advisory(stage):
    if stage == "raw":
        print("Characteristics: Firm, hard, and predominantly green. High in resistant starch, low in sugar.")
        print("Ripening Green bananas in boxes and or cushioned plastic crates should be loaded into the ripening room (lower te
        print("Uses: Ideal for savory dishes, boiling, frying, or making banana chips. Often used as plantains.")

    elif stage == "semi_ripe":
        print("Characteristics: Green tips with a yellow body or fully yellow but firm. Starch is converting to sugar.")
        print("Uses: Good for stir-fries, light snacks, or eating fresh if a less sweet, firmer texture is preferred.")

    elif stage == "ripe":
        print("Characteristics: Fully yellow with brown sugar spots. Soft texture, high sugar content, aromatic.")
        print("Uses: Best for eating fresh as a snack, in fruit salads, or smoothies.")

    elif stage == "over_ripe":
        print("Characteristics: Almost entirely brown/dark brown. Extremely soft and very sweet.")
        print("Uses: Excellent for baking (banana bread, muffins) and blending into smoothies or pancakes due to high sugar cond
```

```
# 🔍 Prediction Function
def predict_image(img_path):
    img = load_img(img_path, target_size=(128,128))
    img_array = img_to_array(img)/255.0
    img_array = np.expand_dims(img_array, axis=0)

    pred = model.predict(img_array)
    idx = np.argmax(pred)
    confidence = np.max(pred)

    stage = class_names[idx]

    print("Stage:", stage)
    print("Confidence:", round(float(confidence), 3))
    print("Advice:", advisory(stage))
```

```
from google.colab import files

uploaded = files.upload()

for file in uploaded.keys():
    predict_image(file)
```

Choose files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
Saving 20230214_092934.jpg to 20230214_092934.jpg
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 81ms/step
Stage: semi_ripe
Confidence: 0.96
Characteristics: Green tips with a yellow body or fully yellow but firm. Starch is converting to sugar.
Uses: Good for stir-fries, light snacks, or eating fresh if a less sweet, firmer texture is preferred.
Advice: None
```